

# Yingrui Hou

## Data Scientist & Ingénieur Modélisation | Validation, Monitoring & Systèmes Industriels

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### SUMMARY

Data scientist et ingénieur en modélisation titulaire d'un doctorat en physique expérimentale, avec plusieurs années d'expérience dans le développement de workflows Python/C++ pour l'analyse de données complexes, la calibration, la validation et le monitoring de systèmes physiques.

Je souhaite transférer cette expérience vers des projets industriels à impact dans les secteurs New Energies & Infrastructures et Connectivity & Data, notamment pour l'analyse de systèmes énergétiques, la maintenance prédictive, la corrélation simulation-essais et le diagnostic d'anomalies.

### CORE SKILLS

- Industrial data and ML: Python, pandas, NumPy, scikit-learn, statsmodels, CatBoost / Gradient Boosting
- Physical modelling: simulation-to-measurement alignment, response modelling, signal analysis, Geant4, ROOT / RooFit
- Validation and monitoring: calibration, model validation, anomaly detection, dataset-shift diagnostics, root-cause analysis
- Statistical methods: probabilistic inference, maximum-likelihood estimation, bias correction, uncertainty-aware analysis
- Engineering workflow: C/C++, Linux, Git, CI/CD practices, YAML / Shell / CMake automation, shared codebases
- Technical delivery: reproducible pipelines, documented results, multidisciplinary collaboration, decision-facing analysis

### PROFESSIONAL EXPERIENCE

#### Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Developed reusable Python/C++ workflows that connected simulations, complex measurement data, and statistical models for decision-facing technical analyses.
- Built monitoring and diagnostic pipelines to compare expected and observed behaviour, detect anomalies and distribution shifts, and support root-cause analysis.
- Implemented calibration and bias-correction methods that improved model credibility when direct ground truth was incomplete and operating conditions differed between datasets.
- Developed CatBoost-based workflows with feature studies, train/test diagnostics, reproducible model application, and dataset-shift mitigation through reweighting.
- Combined experimental understanding, test data, response maps, and simulation outputs to quantify performance limitations and separate physical effects from modelling artefacts.
- Worked in Linux-based shared codebases using Git, modular programming, collaborative review, reproducible configurations, and automated workflows across international teams.
- Translated validation findings into concrete follow-up actions, including calibration updates and the identification of a software issue that was subsequently corrected downstream.

#### Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed Geant4/C++ simulation, signal-modelling, calibration, and validation workflows to connect sensor response, experimental measurements, and system-level performance.

## SELECTED OUTCOMES

- Improved reconstructed-energy resolution by about 2x through time-resolved signal analysis, statistical fitting, and calibration on realistic measurement data.
- Quantified that response non-uniformity contributed less than 1% to system-resolution degradation in a measurement-to-model validation study.

## SELECTED PROJECTS

### Sensor Simulation & Reconstruction

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor response.

*Methods: Geant4 simulation, beam-test analysis, response-map reconstruction*

### Test-Data Validation & Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

*Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics*

### Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

*Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting*

### Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

*Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting*

### Model Correction Workflow

Developed simulation-correction tooling that turns measured physics constraints into an event-level resampling workflow.

*Methods: Gaudi-package development, event-level correction, framework integration*

## EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2